

//SIT55 : ASSIGNMENT 6: EXCEPTION HANDLING

/*Implement a program to handle Arithmetic exception, Array Index Out of Bounds. The user enters two numbers Num1 and Num2. The division of Num1 and Num2 is displayed. If Num1 and Num2 are not integers, the program would throw a Number Format Exception. If Num2 were zero, the program would throw an Arithmetic Exception. Display the exception.*/

```
package Javaprogs;
import java.util.*;
public class ExceptionHandling
{
    public static void main (String args[])
    {
        int num1,num2,result;
        Scanner sc=new Scanner(System.in);
        /* try-catch blocks allows you to define a block of codes to be tested
        for exceptions while it is being executed*/

        try
        {
            int Arr[]={21,66,125,333};
            System.out.println("The value of array element at Index 2 is: "
                               +Arr[2]);
            System.out.println("The value of array element at Index 10 is: "
                               +Arr[10]);
        }
        catch (ArrayIndexOutOfBoundsException e)
        {
            System.out.println(e);
        }
        try
        {
            System.out.println("\nEnter the first number:");
            num1=sc.nextInt();
            System.out.println("Enter the second number:");
            num2=sc.nextInt();
            result=num1/num2;
            System.out.println("The result is: " +result);
        }
        catch (ArithmeticException e)
        {
            System.out.println (e);
        }
        try
        {
            System.out.println("\nEnter the number:");
            String str1=sc.next();
            int a=Integer.parseInt(str1);
            System.out.println("Entered Number is: " +a);
        }
        catch (NumberFormatException e)
        {
            System.out.println (e);
        }
    }
}
```

```
finally //executed regardless of exception occurred or not
{
System.out.println("\n finally BLOCK IS ALWAYS EXECUTED");
}
}
```

/*OUTPUT:

The value of array element at Index 2 is: 125

java.lang.ArrayIndexOutOfBoundsException: Index 10 out of bounds for length 4

ENTER THE FIRST NUMBER:

6

ENTER THE SECOND NUMBER:

0

java.lang.ArithmeticException: / by zero

ENTER THE NUMBER:

abc

java.lang.NumberFormatException: For input string: "abc"

finally BLOCK IS ALWAYS EXECUTED

*/